Ejercicio 3.3: Imputación múltiple

Silvia Pineda

Carga de Datos y Librerías

```r
library(naniar)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

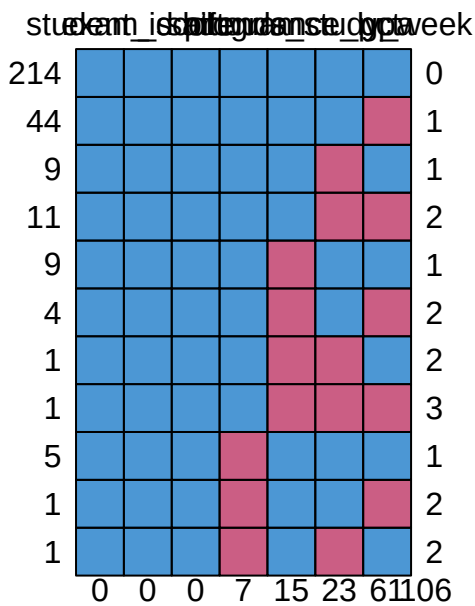
```r
library(rio)

data <- read.csv("students_FP.csv",
  na.strings = c("", "NA", "NaN", "NULL"),
  stringsAsFactors = TRUE
)
```

Imputación múltiple

```r
library(mice)
```

```
Attaching package: 'mice'

The following object is masked from 'package:stats':

    filter

The following objects are masked from 'package:base':

    cbind, rbind
```

```r
md.pattern(data)
```




```
    student_id exam_score shift program attendance_pct hours_study_week gpa
214          1          1     1       1              1                 1   1   0
44           1          1     1       1              1                 1   0   1
9            1          1     1       1              1                 0   1   1
11           1          1     1       1              1                 0   0   2
9            1          1     1       1              0                 1   1   1
4            1          1     1       1              0                 1   0   2
1            1          1     1       1              0                 0   1   2
```

```
1            1          1     1       1              0                 0   0   3
5            1          1     1       0              1                 1   1   1
1            1          1     1       0              1                 1   0   2
1            1          1     1       0              1                 0   1   2
             0          0     0       7             15                23  61 106
```

No hay ninguna observación con todas las variables missing.

```r
impData <- mice(select(data,-student_id),m=5,maxit=50,seed=500)
```

```
 iter imp variable
  1   1  hours_study_week  attendance_pct  gpa  program
  1   2  hours_study_week  attendance_pct  gpa  program
  1   3  hours_study_week  attendance_pct  gpa  program
  1   4  hours_study_week  attendance_pct  gpa  program
  1   5  hours_study_week  attendance_pct  gpa  program
  2   1  hours_study_week  attendance_pct  gpa  program
  2   2  hours_study_week  attendance_pct  gpa  program
  2   3  hours_study_week  attendance_pct  gpa  program
  2   4  hours_study_week  attendance_pct  gpa  program
  2   5  hours_study_week  attendance_pct  gpa  program
  3   1  hours_study_week  attendance_pct  gpa  program
  3   2  hours_study_week  attendance_pct  gpa  program
  3   3  hours_study_week  attendance_pct  gpa  program
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  4   4  hours_study_week  attendance_pct  gpa  program
  4   5  hours_study_week  attendance_pct  gpa  program
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  6   1  hours_study_week  attendance_pct  gpa  program
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  6   5  hours_study_week  attendance_pct  gpa  program
```

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  50   1  hours_study_week  attendance_pct  gpa  program
  50   2  hours_study_week  attendance_pct  gpa  program
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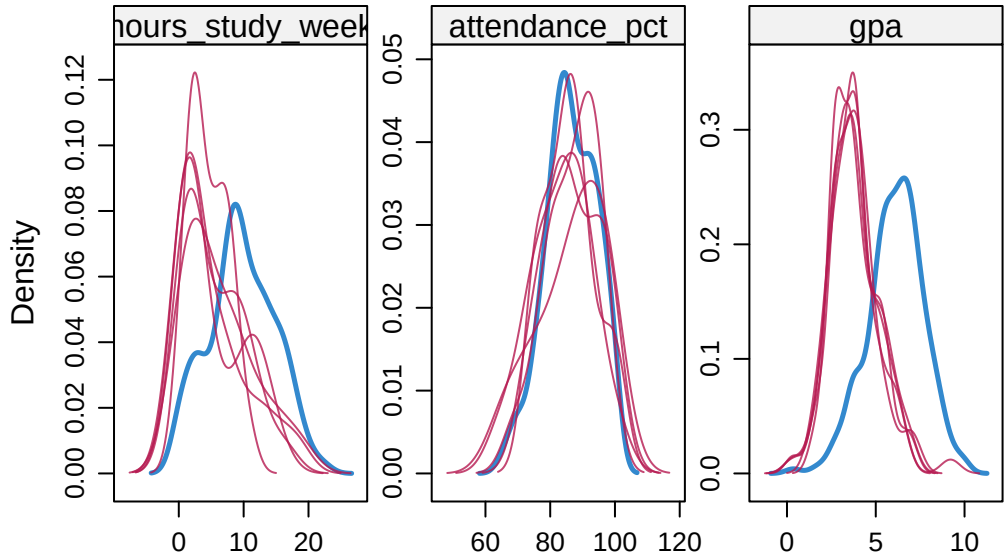
  50   4  hours_study_week  attendance_pct  gpa  program
  50   5  hours_study_week  attendance_pct  gpa  program
```

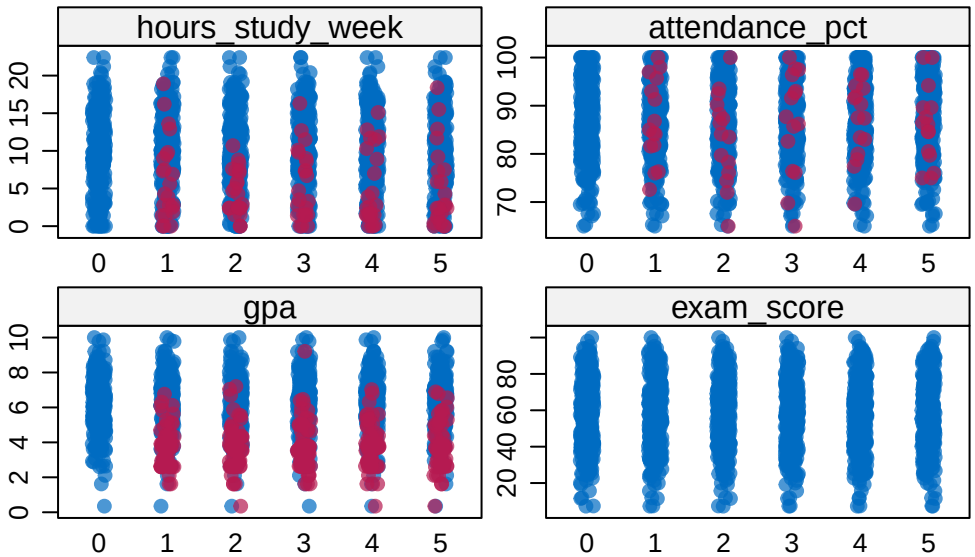
```r
summary(impData)
```

```
Class: mids
Number of multiple imputations:  5
Imputation methods:
hours_study_week   attendance_pct              gpa       exam_score
           "pmm"            "pmm"            "pmm"               ""
         program            shift
       "polyreg"               ""
PredictorMatrix:
                 hours_study_week attendance_pct gpa exam_score program shift
hours_study_week                0              1   1          1       1     1
attendance_pct                  1              0   1          1       1     1
gpa                             1              1   0          1       1     1
exam_score                      1              1   1          0       1     1
program                         1              1   1          1       0     1
shift                           1              1   1          1       1     0
```

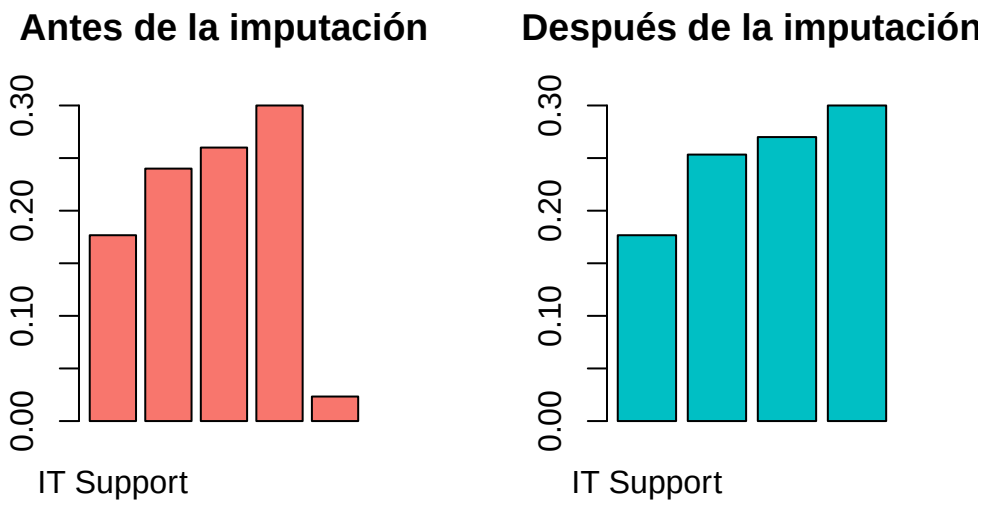
```r
###Visualizar las variables cuantitativas
densityplot(impData)
```



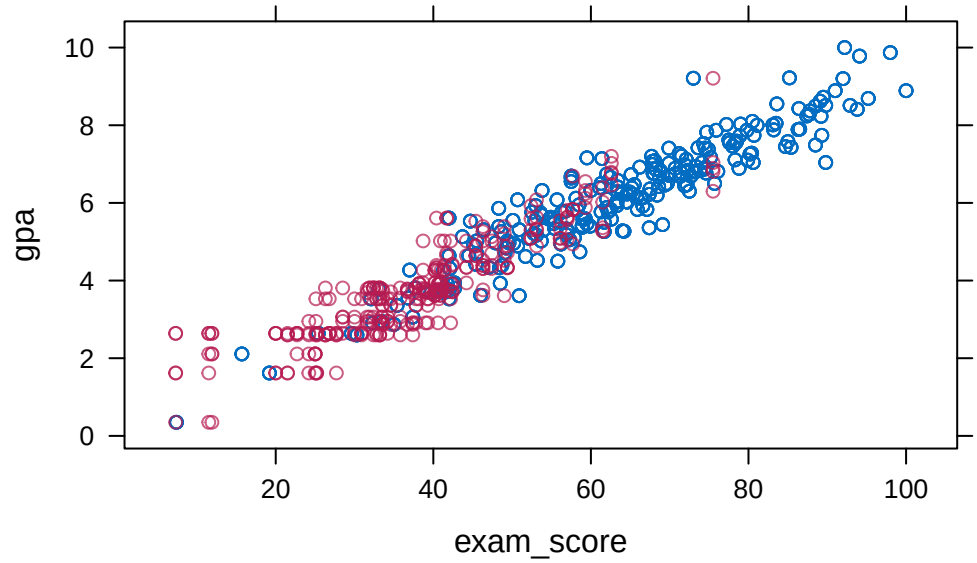

```r
stripplot(impData, pch = 20, cex = 1.2)
```

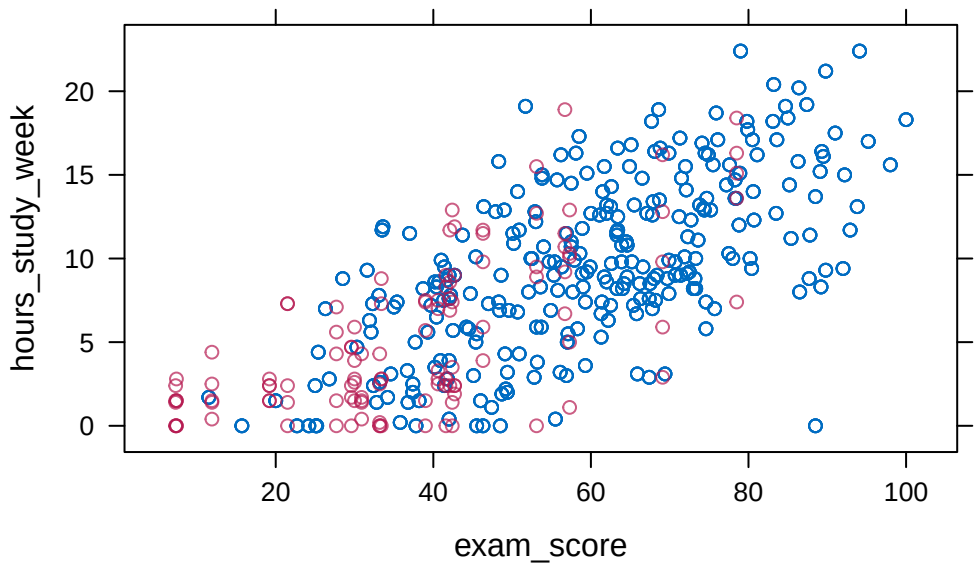


```r
###Visualizar las variables cualitativas
completedData <- complete(impData,1)
par(mfrow = c(1, 2))
barplot(prop.table(table(data$program, useNA = "ifany")),
        main = "Antes de la imputación", col = "#F8766D" )
barplot(prop.table(table(completedData$program)),
        main = "Después de la imputación",col = "#00BFC4")
```




```r
xyplot(impData,gpa~exam_score)
```




```r
xyplot(impData,hours_study_week~exam_score)
```